

1-ma'ruza. Parametrga bog'liq integrallar(6 soat)

Reja:

1. Chegaralari o'zgaruvchi parametrga bog'liq integrallar
2. Parametrga bog'liq xosmas integrallar
3. Parametrga bog'liq xosmas integrallarning parametr bo'yicha tekis yaqinlashish alomatlari.
4. Parametrga bog'liq xosmas integrallarning funksional xossalari.
5. Ba'zi xosmas integrallarni hisoblash, Eyler integrallari.

Faraz qilaylik, $f(x, y)$ funksiya $M_0 = \{(x, y) \in R^2 : a \leq x \leq b, c \leq y \leq d\}$

to'plamda berilgan va har bir tayin $y \in [c, d]$ da $f(x, y)$ funksiya x o'zgaruvchining funksiyasi sifatida $[a, b]$ da integrallanuvchi bo'lsin. $\alpha(y)$ va $\beta(y)$ funksiyalarning har biri $[c, d]$ da berilgan va $\forall y \in [c, d]$ uchun

$$a \leq \alpha(y) \leq \beta(y) \leq b \quad (1)$$

tengsizliklar bajarilsin.

Ushbu

$$\int_{\alpha(y)}^{\beta(y)} f(x, y) dx$$

integral, ravshanki, y o'zgaruvchiga bog'liq bo'ladi:

$$J_1(y) = \int_{\alpha(y)}^{\beta(y)} f(x, y) dx \quad (2)$$

(2) integral chegaralari ham parametrga bog'liq integral deyiladi.

1^o. $J_1(y)$ funksiyaning uzluksizligi. $J_1(y)$ funksiyaning uzluksizligini quyidagi teorema ifodalaydi:

1-teorema. Faraz qilaylik, $f(x, y)$ funksiya M_0 to'plamda uzluksiz bo'lib, $\alpha(y)$ va $\beta(y)$ funksiyalar esa $[c, d]$ segmentda uzluksiz bo'lsin. U holda

$$J_1(y) = \int_{\alpha(y)}^{\beta(y)} f(x, y) dx$$

funksiya $[c, d]$ da uzluksiz bo'ladi.

◀ Ixtiyoriy $y_0 \in [c, d]$ nuqtani olaylik. Integralning ma'lum xossalaridan foydalanib topamiz:

$$\begin{aligned} J_1(y) &= \int_{\alpha(y)}^{\beta(y)} f(x, y) dx = \int_{\alpha(y)}^{\alpha(y_0)} f(x, y) dx + \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y) dx + \int_{\beta(y_0)}^{\beta(y)} f(x, y) dx = \\ &= \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y) dx + \int_{\beta(y_0)}^{\beta(y)} f(x, y) dx - \int_{\alpha(y_0)}^{\alpha(y)} f(x, y) dx. \end{aligned} \quad (3)$$

Ravshanki,

$$\int_{\alpha(y_0)}^{\beta(y_0)} f(x, y) dx$$

integral chegarasi o'zgarmas bo'lgan parametrga bog'liq integral. Bu funksiya 75-ma'ruzada keltirilgan 2-teoremaga muvofiq $\mathcal{Y}$ o'zgaruvchining uzluksiz funksiyasi bo'ladi. Demak,

$$y \rightarrow y_0 \text{ da } \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y) dx \rightarrow \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y_0) dx = J_1(y_0) \quad (4)$$

bo'ladi.

$f(x, y)$ funksiya M_0 to'plamda uzluksiz bo'lganligi sababli $\mathcal{Y}$ shu to'plamda chegaralangan bo'ladi:

$$|f(x, y)| \leq C \quad (C = \text{const}).$$

Shartga ko'ra $\alpha(y)$ va $\beta(y)$ funksiyalar $[c, d]$ segmentda uzluksiz.

Demak,

$$y \rightarrow y_0 \text{ da } \alpha(y) \rightarrow \alpha(y_0),$$

$$y \rightarrow y_0 \text{ da } \beta(y) \rightarrow \beta(y_0).$$

Endi

$$\left| \int_{\beta(y_0)}^{\beta(y)} f(x, y) dx \right| \leq C \left| \beta(y) - \beta(y_0) \right|, \quad \left| \int_{\alpha(y_0)}^{\alpha(y)} f(x, y) dx \right| \leq C \left| \alpha(y) - \alpha(y_0) \right|,$$

munosabatlardan

$$y \rightarrow y_0 \text{ da } \int_{\beta(y_0)}^{\beta(y)} f(x, y) dx \rightarrow 0, \quad (5)$$

$$y \rightarrow y_0 \text{ da } \int_{\alpha(y_0)}^{\beta(y)} f(x, y) dx \rightarrow 0$$

bo'lishini topamiz.

(3) tenglikda, $y \rightarrow y_0$ da limitga o'tish va unda (4) va (5) munosabatlarni hisobga olish natijasida $y \rightarrow y_0$ da $J_1(y) \rightarrow J_1(y_0)$ bo'lishi kelib chiqadi. Demak, $J_1(y)$ funksiya $[c, d]$ da uzluksiz. ►

2°. $J_1(y)$ **funksiyani differentsiallash.** Faraz qilaylik, $f(x, y)$ funksiya

$$M_0 = \{(x, y) \in R^2 : a \leq x \leq b, c \leq y \leq d\}$$

to'plamda, $\alpha(y)$ va $\beta(y)$ funksiyalar esa $[c, d]$ segmentda berilgan bo'lib, $\alpha(y)$, $\beta(y)$ funksiyalar (1) shartni bajarsin, yahni $\forall y \in [c, d]$ uchun

$$a \leq \alpha(y) \leq \beta(y) \leq b$$

bo'lsin.

2-teorema. Aytaylik, $f(x, y)$, $\alpha(y)$ va $\beta(y)$ funksiyalar quyidagi shartlarni bajarsin:

- 1) $f(x, y)$ funksiya M_0 to'plamda uzluksiz;
- 2) $f(x, y)$ funksiya M_0 to'plamda uzluksiz $f'_y(x, y)$ xususiy hosilaga ega;
- 3) $\alpha(y)$ va $\beta(y)$ funksiyalar $[c, d]$ da $\alpha'(y)$ va $\beta'(y)$ hosilalarga ega.

U holda $J_1(y) = \int_{\alpha(y)}^{\beta(y)} f(x, y) dx$ funksiya $[c, d]$ segmentda $J'_1(y)$ hosilaga ega bo'lib,

$$J'_1(y) = \int_{\alpha(y)}^{\beta(y)} f'_y(x, y) dx + \beta'(y) f(\beta(y), y) - \alpha'(y) f(\alpha(y), y)$$

bo'ladi.

◀ $y_0 \in [c, d]$, $y_0 + \Delta y \in [c, d]$ nuqtalarni olib, topamiz:

$$\frac{J_1(y_0 + \Delta y) - J_1(y_0)}{\Delta y} = \frac{1}{\Delta y} \left[\int_{\alpha(y_0 + \Delta y)}^{\beta(y_0 + \Delta y)} f(x, y_0 + \Delta y) dx - \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y_0) dx \right].$$

Agar

$$\begin{aligned} & \int_{\alpha(y_0+\Delta y)}^{\beta(y_0+\Delta y)} f(x, y_0+\Delta y) dx = \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y_0+\Delta y) dx + \\ & + \int_{\beta(y_0)}^{\beta(y_0+\Delta y)} f(x, y_0+\Delta y) dx - \int_{\alpha(y_0)}^{\alpha(y_0+\Delta y)} f(x, y_0+\Delta y) dx \end{aligned}$$

bo'lishini ehtiborga olsak, unda

$$\begin{aligned} \frac{J_1(y_0+\Delta y) - J_1(y_0)}{\Delta y} &= \int_{\alpha(y_0)}^{\beta(y_0)} \frac{f(x, y_0+\Delta y) - f(x, y_0)}{\Delta y} dx + \\ &+ \frac{1}{\Delta y} \int_{\beta(y_0)}^{\beta(y_0+\Delta y)} f(x, y_0+\Delta y) dx - \frac{1}{\Delta y} \int_{\alpha(y_0)}^{\alpha(y_0+\Delta y)} f(x, y_0+\Delta y) dx \end{aligned} \quad (6)$$

bo'lishi kelib chiqadi.

1- teoreмага ko'ra

$$\begin{aligned} \lim_{\Delta y \rightarrow 0} \int_{\alpha(y_0)}^{\beta(y_0)} \frac{f(x, y_0+\Delta y) - f(x, y_0)}{\Delta y} dx &= \\ &= \int_{\alpha(y_0)}^{\beta(y_0)} \lim_{\Delta y \rightarrow 0} \frac{f(x, y_0+\Delta y) - f(x, y_0)}{\Delta y} dx = \int_{\alpha(y_0)}^{\beta(y_0)} f'_y(x, y_0) dx \end{aligned} \quad (7)$$

bo'ladi.

O'rta qiymat haqidagi teoremadan foydalanib, topamiz:

$$\begin{aligned} \int_{\alpha(y_0)}^{\beta(y_0)} f(x, y_0+\Delta y) dx &= f(x', y_0+\Delta y) [\beta(y_0+\Delta y) - \beta(y_0)], \\ \int_{\alpha(y_0)}^{\alpha(y_0+\Delta y)} f(x, y_0+\Delta y) dx &= f(x'', y_0+\Delta y) [\alpha(y_0+\Delta y) - \alpha(y_0)]. \end{aligned}$$

Bunda x' nuqta $\beta(y_0)$, $\beta(y_0+\Delta y)$ nuqtalar orasida, x'' esa $\alpha(y_0)$, $\alpha(y_0+\Delta y)$ nuqtalar orasida joylashgan. $y \rightarrow y_0$ da limitga o'tishi bilan quyidagi tengliklarga kelimiz:

$$\begin{aligned} \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta y} \int_{\beta(y_0)}^{\beta(y_0+\Delta y)} f(x, y_0+\Delta y) dx &= \lim_{\Delta y \rightarrow 0} f(x', y_0+\Delta y) \frac{[\beta(y_0+\Delta y) - \beta(y_0)]}{\Delta y} = \\ &= f(\beta(y_0), y_0) \beta(y_0), \\ \lim_{\Delta y \rightarrow 0} \frac{1}{\Delta y} \int_{\alpha(y_0)}^{\alpha(y_0+\Delta y)} f(x, y_0+\Delta y) dx &= \lim_{\Delta y \rightarrow 0} f(x'', y_0+\Delta y) \frac{[\alpha(y_0+\Delta y) - \alpha(y_0)]}{\Delta y} = \\ &= f(\alpha(y_0), y_0) \alpha(y_0). \end{aligned} \quad (8)$$

Yuqoridagi (6) munosabatda $\Delta y \rightarrow 0$ da limitga o'tib, (7) va (8) tengliklarni ehtiborga olib, ushbu

$$\lim_{\Delta y \rightarrow 0} \frac{J_1(y_0 + \Delta y) - J_1(y_0)}{\Delta y} = \int_{\alpha(y_0)}^{\beta(y_0)} f'_y(x, y_0) dx + f(\beta(y_0), y_0) \beta'(y_0) - f(\alpha(y_0), y_0) \alpha'(y_0)$$

tenglikka kelamiz.

$$J'_1(y_0) = \int_{\alpha(y_0)}^{\beta(y_0)} f'_y(x, y_0) dx + f(\beta(y_0), y_0) \beta'(y_0) - f(\alpha(y_0), y_0) \alpha'(y_0)$$

Demak, ▶

Misol. Ushbu $J_1(y) = \int_0^1 e^x |y - x| dx$ funksiyaning hosilasi topilsin.

◀ Aytaylik, $y \in (-\infty, 0]$ bo'lsin. Bu holda

$$J_1(y) = -y \int_0^1 e^x dx + \int_0^1 x e^x dx = y(1 - e) + [e - (e - 1)] = (1 - e)y + 1$$

bo'lib, $J'_1(y) = 1 - e$ bo'ladi.

Aytaylik, $y \in (0, 1)$ bo'lsin. Bu holda

$$\begin{aligned} J_1(y) &= \int_0^y (y - x) e^x dx - \int_y^1 (y - x) e^x dx = \int_0^y y e^x dx - \\ &- \int_0^y x e^x dx - \int_y^1 y e^x dx + \int_y^1 x e^x dx = 2e^y - (e + 1)y - 1 \end{aligned}$$

bo'lib, $J'_1(y) = 2e^y - e - 1$ bo'ladi.

Aytaylik, $y \in [1, +\infty)$ bo'lsin. Bu holda

$$J_1(y) = y \int_0^1 e^x dx - \int_0^1 x e^x dx = y(e - 1) - [e - (e - 1)] = (e - 1)y - 1$$

bo'lib, $J'_1(y) = e - 1$ bo'ladi.

$$J'_1(y) = \begin{cases} 1 - e, & \text{arab } -\infty < y \leq 0, \\ 2e^y - e - 1, & \text{arab } 0 < y < 1, \\ e - 1, & \text{arab } y \geq 1 \end{cases}$$

Demak,

bo'ladi. ▶

Mashqlar

1. Agar $J_1(y) = \int_0^y \frac{e^x dx}{\sqrt{y-x}}$ ($0 \leq x \leq 1$) bo'lsa, u holda $0 < y < 1$ uchun

$$J_1'(y) = \frac{1}{\sqrt{y}} + \int_0^y \frac{e^x dx}{\sqrt{y-x}}$$
 bo'lishini isbotlansin.

2. Agar $J_1(y) = \int_y^{1+y} \frac{dx}{1+x^2+y^2}$ bo'lsa, $\lim_{y \rightarrow 0} J_1(y)$ topilsin.

Parametrga bog'liq xosmas integrallar

1^o. Parametrga bog'liq xosmas integral tushunchasi. Faraz qilaylik, $f(x, y)$ funksiya

$$M = \{(x, y) \in R^2 : x \in [a, +\infty), y \in E \subset R\}$$

to'plamda berilgan bo'lsin. Bu funksiya har bir tayin $y \in E$ da x o'zgaruvchining funksiyasi sifatida $[a, +\infty)$ da integrallanuvchi, ya'ni

$$\int_a^{+\infty} f(x, y) dx$$

xosmas integral yaqinlashuvchi. Ravshanki, integralning qiymati y o'zgaruvchiga bog'liq bo'ladi:

$$F(y) = \int_a^{+\infty} f(x, y) dx \quad (1)$$

Masalan, $y > 1$ bo'lganda

$$\int_1^{+\infty} \frac{dx}{x^y} = \lim_{t \rightarrow \infty} \int_1^t \frac{dx}{x^y} = \lim_{t \rightarrow \infty} \frac{1}{1-y} (t^{1-y} - 1) = \frac{1}{y-1}$$

bo'ladi. Demak, bu holda

$$F(y) = \frac{1}{y-1}$$

bo'ladi.

(1) integral parametrga bog'liq chegarasi cheksiz xosmas integral, y esa parametr deyiladi.

Xuddi shunga o'xshash

$$F_1(y) = \int_{-\infty}^a f(x, y) dx, \quad F_2(y) = \int_{-\infty}^{+\infty} f(x, y) dx$$

parametrga bog'liq xosmas integrallar tushunchalari kiritiladi.

Aytaylik, $f(x, y)$ funksiya

$$M = \{(x, y) \in R^2 : x \in [a, b), y \in E \subset R\}$$

to'plamda berilgan bo'lsin. Bu funksiya har bir tayin $y \in E$ da x o'zgaruvchining funksiyasi sifatida qaralganda uning uchun b maxsus nuqta bo'lib, u $[a, b)$ da integrallanuvchi, ya'ni

$$\int_a^b f(x, y) dx$$

xosmas integral yaqinlashuvchi bo'lsin. Ravshanki, bu holda ham integralning qiymati y o'zgaruvchiga bog'liq bo'ladi:

$$\Phi(y) = \int_a^b f(x, y) dx \quad (2)$$

Masalan, $0 < y < 1$ bo'lganda

$$\int_1^2 \frac{dx}{(2-x)^y} = \lim_{t \rightarrow 2-0} \int_1^t (2-x)^{-y} dx = \lim_{t \rightarrow 2-0} \frac{1}{y-1} [(2-t)^{1-y} - 1] = \frac{1}{1-y}$$

bo'ladi. Demak, bu holda

$$\Phi(y) = \frac{1}{1-y}$$

bo'ladi.

(2) integral parametrga bog'liq, chegaralanmagan funksiyaning xosmas integrali, y esa parametr deyiladi.

Umumiy holda, parametrga bog'liq, chegaralanmagan funksiyaning chegarasi cheksiz integrali tushunchasi ham yuqoridagidek kiritiladi.

Parametrga bog'liq xosmas integrallarning funksional xossalari (limiti, uzluksizligi, differensiallanishi integrallanishi)ni

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral uchun keltirish bilan kifoyalanamiz.

2^o. Integralning tekis yaqinlashishi. Aytaylik, $f(x, y)$ funksiya

$$M = \{(x, y) \in R^2 : x \in [a, +\infty), y \in E \subset R\}$$

to'plamda berilgan bo'lib, har bir tayin $y \in E$ da

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

xosmas integral yaqinlashuvchi bo'lsin. Ta'rifga binoan

$$F(y) = \int_a^{+\infty} f(x, y) dx = \lim_{t \rightarrow +\infty} \int_a^t f(x, y) dx \quad (a < t < \infty)$$

bo'ladi.

Natijada berilgan $f(x, y)$ funksiya yordamida

$$G(y, t) = \int_a^t f(x, y) dx, \quad (a < t < \infty)$$

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

funksiyalar yuzaga keladi va

$$\lim_{t \rightarrow +\infty} G(y, t) = F(y) \quad (y \in E)$$

munosabat bajariladi.

Demak, $G(y, t)$ funksiya $t \rightarrow +\infty$ da limit funksiya $F(y)$ ga ega bo'ladi.

1-ta'rif. Agar $t \rightarrow +\infty$ da $G(y, t)$ funksiya limit funksiya $F(y)$ ga E to'plamda tekis yaqinlashsa,

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral E to'plamda tekis yaqinlashuvchi deyiladi.

Integralning E to'plamda tekis yaqinlashuvchiligini quyidagicha anglash lozim:

- 1) har bir tayin $y \in E$ da $\int_a^{+\infty} f(x, y) dx$ xosmas integral yaqinlashuvchi;
- 2) $\forall \varepsilon > 0$ olinganda ham, shunday $\delta = \delta(\varepsilon) > 0$ topiladiki, $\forall t > \delta$ va $\forall y \in E$ uchun

$$\left| \int_t^{+\infty} f(x, y) dx \right| < \varepsilon$$

tengsizligi bajariladi.

1-misol. Ushbu

$$\int_0^{+\infty} e^{-x} \cos xy dx$$

xosmas integralning $(-\infty, +\infty)$ da tekis yaqinlashuvchi ekani ko'rsatilsin.

◀ Har bir tayin $y \in (-\infty, +\infty)$ da qarayotgan xosmas integralning yaqinlashuvchi ekanligi ravshan.

$\forall \varepsilon > 0$ ga ko'ra $\delta = \ln \frac{2}{\varepsilon}$ deyilsa, unda $\forall t > \delta$ va $\forall y \in (-\infty, +\infty)$ uchun

$$\left| \int_t^{+\infty} e^{-x} \cos xy dx \right| \leq \int_t^{+\infty} e^{-x} dx = e^{-t} \leq e^{-\delta} = e^{-\ln \frac{2}{\varepsilon}} = \frac{\varepsilon}{2} < \varepsilon$$

bo'ladi. Demak, berilgan integral $(-\infty, +\infty)$ da tekis yaqinlashuvchi. ▶

2-ta'rif. Agar $t \rightarrow +\infty$ da $G(y, t)$ funksiya limit funksiya $F(y)$ ga E to'plamda tekis yaqinlashmasa,

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral E to'plamda tekis yaqinlashmaydi deyiladi.

Integralning E to'plamda yaqinlashuvchi, ammo uning shu to'plamda tekis yaqinlashmaydi degani quyidagini anglatadi:

1) har bir tayin $y \in E$ da $\int_a^{+\infty} f(x, y) dx$ xosmas integral yaqinlashuvchi;

2) $\forall \delta > 0$ olinganda ham, shunday $\varepsilon_0 > 0$, $y_0 \in E$ va $t_1 > \delta$ bo'lgan $t_1 \in [a, +\infty)$ topiladiki,

$$\left| \int_{t_1}^{+\infty} f(x, y_1) dx \right| \geq \varepsilon_0$$

bo'ladi.

2-misol. Ushbu

$$\int_0^{+\infty} ye^{-xy} dx$$

xosmas integralning $(0, +\infty)$ da tekis yaqinlashmasligi ko'rsatilsin.

◀ Ravshanki,

$$\int_0^{+\infty} ye^{-xy} dx = \lim_{t \rightarrow +\infty} \int_0^t ye^{-xy} dx = \lim_{t \rightarrow +\infty} (1 - e^{-ty}) = 1.$$

Demak, berilgan xosmas integral yaqinlashuvchi. Aytay-lik, $y \in E = (0, +\infty)$ bo'lsin.

Ixtiyoriy musbat δ sonni olaylik. Agar $\varepsilon_0 = \frac{1}{3}$, $t_0 > \delta$ va $y_0 = \frac{1}{t_0}$ deb olsak, u holda

$$\left| \int_{t_0}^{+\infty} y_0 e^{-xy_0} dx \right| = e^{-t_0 y_0} = e^{-1} < \frac{1}{3} = \varepsilon_0$$

bo'ladi. Bu esa $\int_0^{+\infty} ye^{-xy} dx$ integral $E = (0, +\infty)$ da tekis yaqinlashmasligini bildiradi. ▶

Yuqoridagi

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

parametrga bog'liq xosmas integralning parametr $\mathcal{Y}$ bo'yicha E to'plamda tekis yaqinlashishini quyidagicha ham ta'rif-lasa bo'ladi.

3-ta'rif. Agar

$$\lim_{t \rightarrow +\infty} \sup_{y \in E} \left| F(y) - \int_a^t f(x, y) dx \right| = \lim_{t \rightarrow +\infty} \sup_{y \in E} \left| \int_t^{+\infty} f(x, y) dx \right| = 0$$

$(a < t < +\infty)$ bo'lsa,

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

xosmas integral E to'plamda tekis yaqinlashuvchi deyiladi.

3-misol. Ushbu

$$F(y) = \int_1^{+\infty} \frac{dx}{x^y}$$

xosmas integralnig $E = [2, +\infty)$ to'plamda tekis yaqinlashuvchi ekani ko'rsatilsin.

◀ Ravshanki, $1 < t < +\infty$ uchun

$$0 \leq \sup_{y \in [2, +\infty)} \left| \int_t^{+\infty} \frac{dx}{x^y} \right| = \sup_{y \in [2, +\infty)} \frac{1}{(y-1)t^{y-1}} \leq \frac{1}{t}$$

bo'lib,

$$\lim_{t \rightarrow +\infty} \sup_{y \in [2, +\infty)} \left| \int_t^{+\infty} \frac{dx}{x^y} \right| = 0$$

bo'ladi. Demak, berilgan xosmas integral $E = [2, +\infty)$ to'plamda tekis yaqinlashuvchi. ►

Endi integralning tekis yaqinlashishini ifodalovchi teoremani keltiramiz.

1-teorema. Ushbu

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integralning E to'plamda tekis yaqinlashuvchi bo'lishi uchun $\forall \varepsilon > 0$ olinganda ham $\mathcal{Y}$ ga bog'liq bo'lmagan shunday $\delta = \delta(\varepsilon) > 0$ topilib, $t' > \delta$, $t'' > \delta$ tengsizliklarni qanoatlantiruvchi $\forall t', t''$ va $\forall y \in E$ da

$$\left| \int_{t'}^{t''} f(x, y) dx \right| < \varepsilon$$

tengsizlikning bajarilishi zarur va yetarli.

Bu teoremaning isboti ravshan.

3°. Parametr ga bog'liq xosmas integrallarning parametr bo'yicha tekis yaqinlashish alomatlari.

2-teorema (Veyershtrass alomati). Aytaylik, $f(x, y)$ funksiya

$$M = \{(x, y) \in R^2 : x \in [a, +\infty), y \in E \subset R\}$$

to'plamda berilgan va har bir tayin $y \in E$ da $f(x, y)$ funksiya $[a, +\infty)$ da integrallanuvchi bo'lsin.

Agar $[a, +\infty)$ da aniqlangan shunday $\phi(x)$ funksiya topilsaki,

1) $\forall x \in [a, +\infty), \forall y \in E$ uchun $|f(x, y)| \leq \phi(x)$ bo'lsa,

$$\int_a^{+\infty} \phi(x) dx$$

2) ushbu $\int_a^{+\infty} \phi(x) dx$ xosmas integral yaqinlashuvchi bo'lsa, u holda

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral E to'plamda tekis yaqinlashuvchi bo'ladi.

Modomiki, $\int_a^{+\infty} \phi(x) dx$ yaqinlashuvchi ekan, unda $\forall \varepsilon > 0$ olinganda ham, shunday $\delta = \delta(\varepsilon) > 0$ topiladiki, $t' > \delta$, $t'' > \delta$ bo'lganda

$$\left| \int_{t'}^{t''} \phi(x) dx \right| < \varepsilon$$

tengsizlik bajariladi.

Ayni paytda,

$$\left| \int_{t'}^{t''} f(x, y) dx \right| \leq \int_{t'}^{t''} |f(x, y)| dx \leq \int_{t'}^{t''} \phi(x) dx \quad (t' < t'')$$

bo'lganligi sababli

$$\left| \int_{t'}^{t''} f(x, y) dx \right| < \varepsilon$$

bo'ladi. Yuqorida keltirilgan 1-teoremaga muvofiq

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral E to'plamda tekis yaqinlashuvchi bo'ladi. ►

4-misol. Ushbu

$$\int_0^{+\infty} \frac{\cos xy}{1+x^2} dx \quad (y \in E = (-\infty, +\infty))$$

integralning tekis yaqinlashuvchi ekani ko'rsatilsin.

◄ Ravshanki, $\forall x \in [0, +\infty)$ va $\forall y \in (-\infty, +\infty)$ uchun

$$|f(x, y)| = \left| \frac{\cos xy}{1+x^2} \right| \leq \frac{1}{1+x^2}$$

bo'ladi. Ayni paytda,

$$\int_0^{+\infty} \frac{1}{1+x^2} dx$$

xosmas integral yaqinlashuvchi bo'lganligi sababli Veyersh-trass alomatiga ko'ra berilgan integral $E = (-\infty, +\infty)$ da tekis yaqinlashuvchi bo'ladi. ►

Integrallarning tekis yaqinlashishini aniqlashda ko‘p foydalaniladigan Abel hamda Dirixle alomatlarini isbotsiz keltiramiz.

3-teorema (Abel alomati). $f(x, y)$ va $g(x, y)$ funksiyalar

$$M = \{(x, y) \in R^2 : x \in [a, +\infty), y \in E \subset R\}$$

to‘plamda berilgan bo‘lib, quyidagi shartlar bajarilsin:

- 1) har bir tayin $y \in E$ da $g(x, y)$ funksiya $[a, +\infty)$ da monoton bo‘lsin;
- 2) $\forall (x, y) \in M$ uchun $|g(x, y)| \leq c$, ($c = \text{const}$) bo‘lsin;
- 3) ushbu

$$\int_a^{+\infty} f(x, y) dx$$

integral E to‘plamda tekis yaqinlashuvchi bo‘lsin.

U holda

$$\int_a^{+\infty} f(x, y) g(x, y) dx$$

integral E to‘plamda tekis yaqinlashuvchi bo‘ladi.

4-teorema (Dirixle alomati). $f(x, y)$ va $g(x, y)$ funksiyalar M to‘plamda berilgan bo‘lib, quyidagi shartlar bajarilsin:

- 1) $\forall t \geq a$ hamda $\forall t \in E$ da

$$\left| \int_a^t f(x, y) dx \right| \leq c \quad (c = \text{const})$$

tengsizlik bajarilsin;

- 2) har bir tayin $y \in E$ da $g(x, y)$ funksiya limit funksiya $\phi(x) = 0$ ga tekis yaqinlashsin.

U holda

$$\int_a^{+\infty} f(x, y) g(x, y) dx$$

integral E to‘plamda tekis yaqinlashuvchi bo‘ladi.

5-misol. Ushbu

$$\int_0^{+\infty} \frac{\sin xy}{x} dx \quad (y \in E = [1, 2])$$

integral tekis yaqinlashuvchilikka tekshirilsin.

◀ Berilgan integralda

$$f(x, y) = \sin xy, \quad g(x, y) = \frac{1}{x}$$

deyilsa, unda

1) $\forall t > 0, \forall y \in [1, 2]$ uchun

$$\left| \int_0^t f(x, y) dx \right| = \left| \int_0^t \sin xy dx \right| = \left| \frac{1 - \cos ty}{y} \right| \leq 2,$$

2) $x \rightarrow +\infty$ da $g(x, y) = \frac{1}{x}$ funksiya $E = [1, 2]$ da nolga tekis yaqinlashuvchi.

Dirixle alomatiga ko'ra berilgan integral $E = [1, 2]$ da tekis yaqinlashuvchi bo'ladi. ▶

Mashqlar

1. Ushbu

$$\int_0^{+\infty} \frac{y \cos xy^2 dx}{y + x^y}$$

integralning $E = [2, 10]$ to'plamda tekis yaqinlashishi isbotlansin.

2. Ushbu

$$\int_0^{+\infty} \frac{dx}{1 + x^y} \quad (y > 1)$$

integral tekis yaqinlashishga tekshirilsin.

3. Aytaylik, $f(x)$ funksiya R da uzluksiz bo'lib, $\forall x \in R$ da $f(x) \geq 0$ bo'lsin. Ushbu

$$\int_0^{+\infty} f(y-x) dx, \quad \int_{-\infty}^0 f(y-x) dx$$

integrallarning $\mathcal{Y}$ parametr bo'yicha ixtiyoriy chekli $[a, b] \subset R$ segmentda tekis yaqinlashuvchi bo'lishi isbotlansin.

Parametrga bog'liq xosmas integrallarning funksional xossalari

Ushbu ma'ruzada parametrga bog'liq xosmas integral

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

ning limiti, uzluksizligi, differensiallanishi hamda integrallanishi masalalarini bayon etamiz.

1°. $F(y)$ funksiyaning limiti. Aytaylik, $f(x, y)$ funksiya

$$M = \{(x, y) \in R^2 : x \in [a, +\infty), y \in E \subset R\}$$

to'plamda berilgan, $y_0 \in R$ esa Y to'plamning limit nuqtasi bo'lsin.

1-teorema. $f(x, y)$ funksiya qo'yidagi shartlarni bajar-sin:

1) har bir tayin $y \in E$ da $f(x, y)$ funksiya x o'zgaruvchining funksiyasi sifatida $[a, +\infty)$ da uzluksiz;

2) $y \rightarrow y_0$ da $f(x, y)$ funksiya ixtiyoriy $[a, t]$ da ($a < t < \infty$) limit funksiya $\varphi(x)$ ga tekis yaqinlashsin;

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

3) ushbu integral Y to'plamda tekis yaqinlashuvchi bo'lsin. U holda $y \rightarrow y_0$ da $F(y)$ funksiya limitga ega va

$$\lim_{y \rightarrow y_0} F(y) = \int_a^{+\infty} \varphi(x) dx$$

bo'ladi.

◀ Teoremaning 1- va 2- shartlarining bajarilishidan $\varphi(x)$ funksiyaning $[a, \infty)$ da uzluksiz bo'lishini topamiz. Binobarin, $\varphi(x)$ ixtiyoriy $[a, t]$ da ($a < t < \infty$) integrallanuvchi bo'ladi.

Modomiki,

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral E to'plamda tekis yaqinlashuvchi ekan, unda 77-ma'ruzadagi 1-teoremaga ko'ra

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, t' > \delta, t'' > \delta, \forall t', t'', \forall y \in E:$$

$$\left| \int_{t'}^{t''} f(x, y) dx \right| < \varepsilon$$

bo'ladi.

Keyingi tengsizlikda, $y \rightarrow y_0$ da limitga o'tsak, u holda

$$\left| \int_{t'}^{t''} \varphi(x) dx \right| \leq \varepsilon$$

tengsizlik hosil bo'ladi. Bundan $\varphi(x)$ funksiyaning $[a, \infty)$ da integrallanuvchiligi kelib chiqadi.

Ushbu

$$\left| \int_a^{+\infty} f(x, y) dx - \int_a^{+\infty} \varphi(x) dx \right|$$

ayirmani qaraymiz. Uning uchun quyidagi tengsizlik bajariladi:

$$\left| \int_a^{+\infty} f(x, y) dx - \int_a^{+\infty} \varphi(x) dx \right| \leq \int_a^t |f(x, y) - \varphi(x)| dx + \left| \int_t^{+\infty} f(x, y) dx \right| + \left| \int_t^{+\infty} \varphi(x) dx \right|, \quad (a < t < \infty) \quad (1)$$

Bu tengsizlikning o'ng tomonidagi qo'shiluvchilarni baholaymiz. $\int_a^{+\infty} f(x, y) dx$ integral E to'plamda tekis yaqinlashuv-chi bo'lganligi sababli,

$$\forall \varepsilon > 0, \exists \delta_1 = \delta_1(\varepsilon) > 0, \quad \forall t > \delta_1, \quad \forall y \in E$$

$$\left| \int_t^{+\infty} f(x, y) dx \right| < \frac{\varepsilon}{3} \quad (2)$$

bo'ladi.

$$\int_a^{+\infty} \varphi(x) dx$$

integral yaqinlashuvchi bo'lganligi sababli

$$\forall \varepsilon > 0, \exists \delta_2 = \delta_2(\varepsilon) > 0, \quad \forall t > \delta_2 :$$

$$\left| \int_t^{+\infty} \varphi(x) dx \right| < \frac{\varepsilon}{3} \quad (3)$$

bo'ladi.

Ravshanki, $\forall t > \delta_0 : (\delta_0 = \max(\delta_1, \delta_2))$ da (2) va (3) tensizliklar bir yo'la bajariladi. Funktsiya $y \rightarrow y_0$ da $f(x, y)$ $[a, t]$ da $(t > \delta_0)$ limit funksiya $\varphi(x)$ ga tekis yaqinlashuvchi bo'lganligi sababli

$$\forall \varepsilon > 0, \exists \delta' = \delta'(\varepsilon) > 0, |y - y_0| < \delta' \quad \forall y \in E, \forall x \in [a, t] \quad (a < t < \infty)$$

$$|f(x, y) - \varphi(x)| < \frac{\varepsilon}{3(t - a)} \quad (4)$$

bo'ladi.

(1), (2), (3) va (4) munosabatlardan

$$\left| \int_a^{+\infty} f(x, y) dx - \int_a^{+\infty} \varphi(x) dx \right| < \varepsilon$$

bo'lishi kelib chiqadi. Demak

$$\lim_{y \rightarrow y_0} F(y) = \lim_{y \rightarrow y_0} \int_a^{+\infty} f(x, y) dx = \int_a^{+\infty} \varphi(x) dx \quad \blacktriangleright$$

Keyingi tenglikni quyidagicha ham yozish mumkin

$$\lim_{y \rightarrow y_0} \int_a^{+\infty} f(x, y) dx = \int_a^{+\infty} \left[\lim_{y \rightarrow y_0} f(x, y) \right] dx \quad .$$

1- misol. Ushbu

$$\lim_{y \rightarrow +0} \int_a^{+\infty} e^{-xy} \frac{\sin x}{x} dx = \int_a^{+\infty} \frac{\sin x}{x} dx$$

tenglik isbotlansin.

◀ Agar $\varphi(x) = \frac{\sin x}{x}$ funksiyaning $x=0$ nuqtadagi qiymati-ni $\varphi(0)=1$ deb olinsa, unda

$$f(x, y) = e^{-xy} \frac{\sin x}{x}$$

funksiya $\{(x, y) \in R^2 : x \in [0, +\infty), y \in [0, +\infty)\}$ to'plamda uzluksiz bo'ladi.

Ravshanki, har bir tayin $y \in [0, +\infty)$ da $f(x, y)$ funksiya x o'zgaruvchining funksiya sifatida $[0, +\infty)$ da uzluksiz bo'lib, $y \rightarrow +\infty$ da bu funksiya ixtiyoriy $[0, t]$ da

$(0 < t < +\infty)$ $\varphi(x) = \frac{\sin x}{x}$ funksiya ga tekis yaqinlashadi.

Endi,

$$\int_0^{+\infty} e^{-xy} \frac{\sin x}{x} dx$$

xosmas integralni parametr y bo'yicha $[0, +\infty)$ da tekis yaqinlashuvchi bo'lishini ko'rsatamiz.

Agar 77-ma'ruzada keltirilgan Abel alomatida $f(x, y)$ funksiya sifatida $\frac{\sin x}{x}$, $g(x, y)$ funksiya sifatida e^{-xy} funksiyalar olinsa, ular uchun Abel alomatining barcha shartlarining o'rinli bo'lishini ko'rsatish qiyin emas. Demak, alomatga ko'ra

$$\int_0^{+\infty} e^{-xy} \frac{\sin x}{x} dx$$

integral tekis yaqinlashuvchi.

Yuqorida keltirilgan 1-teoremaga binoan

$$\lim_{y \rightarrow +0} \int_0^{+\infty} e^{-xy} \frac{\sin x}{x} dx = \int_0^{+\infty} \left(\lim_{y \rightarrow +0} e^{-xy} \frac{\sin x}{x} \right) dx$$

bo'lib, undan

$$\lim_{y \rightarrow +0} \int_0^{+\infty} e^{-xy} \frac{\sin x}{x} dx = \int_0^{+\infty} \frac{\sin x}{x} dx$$

bo'lishi kelib chiqadi. ►

2^o. $F(y)$ funksiyaning uzluksizligi. Aytaylik, $f(x, y)$ funksiya

$$M_0 = \{(x, y) \in R^2 : x \in [a, +\infty), y \in [c, d]\}$$

to'plamda berilgan bo'lsin.

2-teorema. Agar $f(x, y)$ funksiya M_0 to'plamda uzluksiz bo'lib,

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral $[c, d]$ da tekis yaqinlashuvchi bo'lsa, u holda $F(y)$ funksiya $[c, d]$ da uzluksiz bo'ladi.

◀ Ixtiyoriy $y_0 \in [c, d]$, $y_0 + \Delta y \in [c, d]$ nuqtalarni olib, $F(y)$ funksiyaning orttirmasini topamiz:

$$\Delta F(y_0) = F(y_0 + \Delta y) - F(y_0) = \int_a^{+\infty} [f(x, y_0 + \Delta y) - f(x, y_0)] dx$$

Shartga ko'ra $\int_a^{+\infty} f(x, y) dx$ integral $[c, d]$ da tekis yaqinla-shuvchi. Unda $\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, \forall t_0 > \delta, \forall y \in [c, d]:$

$$\left| \int_{t_0}^{+\infty} f(x, y) dx \right| < \frac{\varepsilon}{3} \quad (5)$$

bo'ladi. Ravshanki, $f(x, y)$ funksiya

$$M_{t_0} = \{(x, y) \in R^2 : x \in [a, t_0], y \in [c, d]\} \quad (a < t_0 < +\infty)$$

to'plamda tekis uzluksiz bo'ladi. Unda

$$\forall \varepsilon > 0, \exists \delta_1 = \delta_1(\varepsilon) > 0, \Delta y < \delta_1(\varepsilon)$$

$$|f(x, y_0 + \Delta y) - f(x, y_0)| < \frac{\varepsilon}{3(t_0 - a)} \quad (6)$$

bo'ladi. Agar $\delta_0 = \max\{\delta, \delta_1\}$ deyilsa uning uchun (5) va (6) tengsizliklar bir yo'la bajariladi. (5) va (6) munosabatlarni e'tiborga olib topamiz:

$$\begin{aligned} |\Delta F(y_0)| &= \left| \int_a^{+\infty} [f(x, y_0 + \Delta y) - f(x, y_0)] dx \right| \leq \\ &\leq \int_a^{t_0} |f(x, y_0 + \Delta y) - f(x, y_0)| dx + \left| \int_{t_0}^{+\infty} f(x, y_0 + \Delta y) dx \right| + \left| \int_{t_0}^{+\infty} f(x, y_0) dx \right| < \varepsilon. \end{aligned}$$

Demak,

$$\lim_{\Delta y \rightarrow 0} \Delta F(y_0) = 0.$$

Bu esa $F(y)$ funksiyaning $[c, d]$ oraliqda uzluksizligini bildiradi. ►

2-misol. Ushbu

$$F(y) = \int_0^{+\infty} e^{-(x-y)^2} dx$$

integral parametr y ning uzluksiz funksiyasi bo'lishi ko'rsatilsin.

◀ Berilgan integralda

$$x - y = t$$

almashtirish bajaramiz. Unda

$$F(y) = \int_{-y}^{+\infty} e^{-t^2} dt = \int_{-y}^0 e^{-t^2} dt + \int_0^{+\infty} e^{-t^2} dt = \int_0^y e^{-t^2} dt + \int_0^{+\infty} e^{-t^2} dt$$

bo'lib, bu yig'indining har bir qo'shiluvchisi y ning uzluksiz funksiyasi bo'lgani uchun berilgan integral parametr y ning uzluksiz funksiyasi bo'ladi. ►

3°. $F(y)$ funksiyani differensiallash. Faraz qilaylik $f(x, y)$ funksiya M_0 to'plamda berilgan bo'lsin.

3-teorema. $f(x, y)$ funksiya quyidagi shartlarni qanoatlantirsin:

- 1) $f(x, y)$ funksiya M_0 to'plamda uzluksiz;
- 2) $f'_y(x, y)$ xususiy hosila mavjud va u M_0 to'plamda uzluksiz;
- 3) Har bir tayin $y \in [c, d]$ da

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral yaqinlashuvchi;

4) Ushbu

$$\int_a^{+\infty} f'_y(x, y) dx$$

integral $[c, d]$ da tekis yaqinlashuvchi.

U holda $F(y)$ funksiya $[c, d]$ da $F'(y)$ hosilaga ega va

$$F'(y) = \int_a^{+\infty} f'_y(x, y) dx$$

bo'ladi.

◀ $y_0 \in [c, d]$, $y_0 + \Delta y \in [c, d]$ nuqtalarni olib, topamiz:

$$\frac{F(y_0 + \Delta y) - F(y_0)}{\Delta y} = \int_a^{+\infty} \frac{f(x, y_0 + \Delta y) - f(x, y_0)}{\Delta y} dx.$$

Lagranj teoremasiga ko'ra

$$\frac{f(x, y_0 + \Delta y) - f(x, y_0)}{\Delta y} = f'_y(x, y_0 + \theta \Delta y), \quad (0 < \theta < 1),$$

$$\frac{F(y_0 + \Delta y) - F(y_0)}{\Delta y} = \int_a^{+\infty} f'_y(x, y_0 + \theta \Delta y) dx$$

bo'ladi. Demak,

$$\lim_{\Delta y \rightarrow 0} \frac{F(y_0 + \Delta y) - F(y_0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \int_a^{+\infty} f'_y(x, y_0 + \theta \Delta y) dx. \quad (7)$$

Shartga ko'ra $f'_y(x, y)$ funksiya M_0 to'plamda uzluksiz. Kantor teoremasiga binoan u

$$M_t = \{(x, y) \in R^2 : x \in [a, t], y \in [c, d]\} \quad (a < t < \infty)$$

to'plamda tekis uzluksiz bo'ladi. Unda

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, |\Delta y| < \delta(\varepsilon) \quad \forall x \in [a, t]$$

$$|f'_y(x, y_0 + \theta \Delta y) - f'_y(x, y_0)| < \varepsilon$$

bo'ladi. Demak $\Delta y \rightarrow 0$ da $f'_y(x, y_0 + \theta \Delta y)$ funksiya $f'_y(x, y_0)$ ga tekis yaqinlashadi. Shartga ko'ra

$$\int_a^{+\infty} f'_y(x, y_0) dx$$

integral tekis yaqinlashuvchi. Unda

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, t' > \delta; t'' > \delta; \quad \forall t', t'', \quad \forall y \in [c, d].$$

$$\left| \int_{t'}^{t''} f'_y(x, y) dx \right| < \varepsilon$$

jumladan,

$$\left| \int_{t'}^{t''} f'_y(x, y_0 + \theta \Delta y) dx \right| < \varepsilon$$

bo'ladi. Keyingi tengsizlikning bajarilishidan esa

$$\int_a^{+\infty} f'_y(x, y_0 + \theta \Delta y) dx$$

integralning tekis yaqinlashuviligi kelib chiqadi. Ushbu ma'ruzada keltirilgan 1-teoremani (7) tenglikning o'ng tomoniga qo'llab, topamiz.

$$\lim_{\Delta y \rightarrow 0} \int_a^{+\infty} f'_y(x, y_0 + \theta \Delta y) dx = \int_a^{+\infty} \lim_{\Delta y \rightarrow 0} f'_y(x, y_0 + \theta \Delta y) dx = \int_a^{+\infty} f'_y(x, y_0) dx. \quad (8)$$

(7) va (8) munosabatlardan

$$F'(y_0) = \int_a^{+\infty} f'_y(x, y_0) dx \quad (9)$$

bo'lishi kelib chiqadi.

(9) munosabatni quyidagicha ham yozish mumkin:

$$\frac{d}{dy} \int_a^{+\infty} f(x, y) dx = \int_a^{+\infty} \frac{\partial f(x, y)}{\partial y} dy.$$

Bu differensiallash amalini integral ostiga o'tkazish qoidasini ifodalaydi. ►

4^o. $F(y)$ funksiyani integrallash. Aytaylik, $f(x, y)$ funksiya

$$M_0 = \{(x, y) \in R^2 : x \in [a, +\infty), y \in [c, d]\}$$

to'plamda berilgan bo'lsin.

4-teorema. Agar $f(x, y)$ funksiya M_0 to'plamda uzluksiz va $F(y) = \int_a^{+\infty} f(x, y) dx$ integral $[c, d]$ da tekis yaqinlashuvchi bo'lsa, u holda $F(y)$ funksiya $[c, d]$ da integrallanuvchi va

$$\int_c^d F(y) dy = \int_c^d \left[\int_a^{+\infty} f(x, y) dx \right] dy = \int_a^{+\infty} \left[\int_c^d f(x, y) dy \right] dx$$

bo'ladi.

◀ Ravshanki, $F(y)$ funksiya $[c, d]$ da uzluksiz bo'ladi. Binobarin, u $[c, d]$ da integrallanuvchi.

Shartga ko'ra

$$F(y) = \int_a^{+\infty} f(x, y) dx$$

integral $[c, d]$ da tekis yaqinlashuvchi. Unda

$$\forall \varepsilon > 0, \exists \delta = \delta(\varepsilon) > 0, \forall t > \delta, \forall y \in [c, d].$$

$$\left| \int_t^{+\infty} f(x, y) dx \right| < \varepsilon$$

bo'ladi. Shu munosabatdagi t ni olib topamiz:

$$\int_a^{+\infty} f(x, y) dx = \int_a^t f(x, y) dx + \int_t^{+\infty} f(x, y) dx.$$

Natijada

$$\begin{aligned} \int_c^d \left[\int_a^{+\infty} f(x, y) dx \right] dy &= \int_c^d \left[\int_a^t f(x, y) dx \right] dy + \int_c^d \left[\int_t^{+\infty} f(x, y) dx \right] dy = \\ &= \int_a^t \left[\int_c^d f(x, y) dy \right] dx + \int_c^d \left[\int_t^{+\infty} f(x, y) dx \right] dy \end{aligned}$$

bo'lad.

Agar

$$\left| \int_c^d F(y) - \int_a^t \left[\int_c^d f(x, y) dy \right] dx \right| \leq \int_c^d \left| \int_t^{+\infty} f(x, y) dx \right| dy < \varepsilon(d - c)$$

bo'lishini e'tiborga olsak, unda

$$\int_c^d F(y) = \lim_{t \rightarrow \infty} \int_a^t \left[\int_c^d f(x, y) dy \right] dx = \int_a^{+\infty} \left[\int_c^d f(x, y) dy \right] dx$$

bo'lib,

$$\int_c^d \left[\int_a^{+\infty} f(x, y) dx \right] dy = \int_a^{+\infty} \left[\int_c^d f(x, y) dy \right] dx$$

ekanligi kelib chiqadi. ►

Mashqlar

1. Agar $f(x)$ funksiya $(0, \infty)$ da integrallanuvchi bo'lib,

$$F(y) = \int_0^{\infty} e^{-yx} f(x) dx$$

bo'lsa, $\lim_{y \rightarrow +0} F(y) = \lim_{y \rightarrow +0} \int_0^{\infty} e^{-yx} f(x) dx = \int_0^{\infty} f(x) dx$ bo'lishi isbotlansin.

2. Ushbu $F(y) = \int_0^{\infty} e^{-y^2x} y dx$ ($-\infty < y < +\infty$) funksiyaning uzluksizlikka tekshirilsin.

Ba'zi xosmas integrallarni hisoblash

Parametrga bog'liq integrallar va ularning funksional xossaligidan foydalanib, ba'zi xosmas integrallarni hisoblaymiz.

1°. $\int_0^{+\infty} \frac{\sin x}{x} dx$ integralni hisoblash. Bu integralning yaqinlashuvchiligi 77-ma'ruzada keltirilgan.

Ma'lumki,

$$\int_0^{+\infty} e^{-xy} \sin x dx = \frac{1}{1+y^2}. \quad (1)$$

Bu tenglikdagi

$$F(y) = \int_0^{+\infty} e^{-xy} \sin x dx$$

parametrga bog'liq integral parametr y bo'yicha ixtiyoriy $[t, A]$ da ($t > 0$) tekis yaqinlashuvchi bo'ladi. Bu tasdiq

$$|e^{-xy} \sin x| < e^{-ty}, \quad \int_0^{+\infty} e^{-ty} dx = \frac{1}{t}$$

bo'lishi hamda Veyershtrass alomatini qo'llashdan kelib chiqadi. (1) tenglikni integrallab topamiz:

$$\int_t^A \left[\int_0^{+\infty} e^{-xy} \sin x dx \right] dy = \int_t^A \frac{1}{1+y^2} dy = \arctg A - \arctg t.$$

Bu tenglikni chap tomonidagi integral uchun

$$\int_t^A \left[\int_0^{+\infty} e^{-xy} \sin x dx \right] dy = \int_0^{+\infty} \left[\int_t^A e^{-xy} \sin x dy \right] dx = \int_0^{+\infty} \frac{e^{-tx} - e^{-Ax}}{x} \sin x dx$$

va $\forall x \geq 0$ da $|\sin x| \leq x$ bo'lib,

$$\left| \int_0^{+\infty} \frac{e^{-Ax}}{x} \sin x dx \right| \leq \int_0^{+\infty} e^{-Ax} dx = \frac{1}{A}$$

bo'ladi. Natijada $A \rightarrow +\infty$ da

$$\frac{\pi}{2} - \arctg t = \int_0^{+\infty} e^{-tx} \frac{\sin x}{x} dx \quad (2)$$

bo'lishi kelib chiqadi.

Endi

$$\lim_{t \rightarrow +0} \int_0^{+\infty} e^{-tx} \frac{\sin x}{x} dx = \int_0^{+\infty} \frac{\sin x}{x} dx$$

tenglikni o'rinli ekanini (qaralsin, 78-ma'ruza) e'tiborga olib (2) da $t \rightarrow +0$ da limitga o'tib topamiz:

$$\int_0^{+\infty} \frac{\sin x}{x} dx = \frac{\pi}{2}.$$

$$\int_0^{+\infty} \frac{\sin yx}{x} dx$$

2^o. 0 **integralni hisoblash.** Bu integralning $(-\infty, +\infty)$ da yaqinlashuvchi bo'lishi ravshan. Aytaylik, $y > 0$ bo'lsin. Bu holda integralda $yx = t$ almashtirish bajarib topamiz:

$$\int_0^{+\infty} \frac{\sin yx}{x} dx = \int_0^{+\infty} \frac{\sin t}{t} dt = \frac{\pi}{2}.$$

Aytaylik, $y < 0$ bo'lsin. Bu holda qaralayotgan integralda $yx = -t$ almashtirish bajarib topamiz:

$$\int_0^{+\infty} \frac{\sin yx}{x} dx = - \int_0^{+\infty} \frac{\sin t}{t} dt = -\frac{\pi}{2}.$$

Aytaylik, $y = 0$ bo'lsin. Bu holda

$$\int_0^{+\infty} \frac{\sin 0x}{x} dx = 0$$

bo'ladi. Demak,

$$\int_0^{+\infty} \frac{\sin yx}{x} dx = \begin{cases} \frac{\pi}{2}, \text{ arap } y > 0 \\ 0, \text{ arap } y = 0 \\ -\frac{\pi}{2}, \text{ arap } y < 0 \end{cases}$$

ya'ni,

$$\int_0^{+\infty} \frac{\sin yx}{x} dx = \frac{\pi}{2} \operatorname{sign} y$$

bo'ladi.

$$\int_0^{+\infty} \frac{x^{a-1}}{1+x} dx$$

3^o. 0 **integralni hisoblash.** Avvalo bu parametrga bog'liq xosmas integralni yaqinlashuvchilikka tekshiramiz. Uning uchun berilgan integralni quyidagicha yozib olamiz:

$$F(a) = \int_0^{+\infty} \frac{x^{a-1}}{1+x} dx = \int_0^1 \frac{x^{a-1}}{1+x} dx + \int_1^{+\infty} \frac{x^{a-1}}{1+x} dx. \quad (3)$$

Aytaylik, $0 < x < 1$ bo'lsin. Bu holda

$$\frac{x^{a-1}}{1+x} < x^{a-1}$$

bo'lib, $a > 0$ da ushbu

$$\int_0^1 x^{a-1} dx$$

integralning yaqinlashuvchi bo'lganligidan, $a > 0$ da

$$\int_0^1 \frac{x^{a-1}}{1+x} dx$$

integralning ham yaqinlashuvchi bo'lishi kelib chiqadi.

Aytaylik, $x \geq 1$ bo'lsin. Bu holda

$$\frac{x^{a-1}}{1+x} < x^{a-2}$$

bo'lib, $a < 1$ da ushbu

$$\int_1^{+\infty} x^{a-2} dx$$

integralning yaqinlashuvchi bo'lganligidan, $a < 1$ da

$$\int_1^{+\infty} \frac{x^{a-1}}{1+x} dx$$

integralning ham yaqinlashuvchi bo'lishi kelib chiqadi.

Demak, qaralayotgan

$$F(a) = \int_0^{+\infty} \frac{x^{a-1}}{1+x} dx$$

integral $0 < a < 1$ da yaqinlashuvchi bo'ladi.

Endi $F(a)$ integralni hisoblaymiz. Ma'lumki, $0 < x < 1$ da

$$\frac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k.$$

Bu tenglikdan

$$\frac{x^{a-1}}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^{a+k-1}$$

bo'lishini topamiz. Tenglikning o'ng tomonidagi qator $[a_0, b_0]$ da $(0 < a_0 \leq x \leq b_0 < 1)$ tekis yaqinlashuvchi bo'lib, uning qisman yig'indisi

$$S_n(x) = \sum_{k=0}^{n-1} (-1)^k x^{a+k-1} = \frac{x^{a-1}(1 - (-x)^n)}{1+x}$$

bo'ladi.

Agar $\forall n \in N, \forall x \in (0, 1)$ uchun

$$\frac{x^{a-1}(1 - (-x)^n)}{1+x} < x^{a-1}$$

tengsizlikni hamda $\int_0^1 x^{a-1} dx$ ($0 < a < 1$) integralning yaqinlashuvchanligini e'tiborga olsak, unda Veyershtrass alomatiga ko'ra

$$\int_0^1 S_n(x) dx$$

tekis yaqinlashuvchi bo'ladi.

Demak,

$$\lim_{n \rightarrow \infty} \int_0^1 S_n(x) dx = \int_0^1 \left(\lim_{n \rightarrow \infty} S_n(x) \right) dx$$

ya'ni,

$$\lim_{n \rightarrow \infty} \int_0^1 \left(\sum_{k=0}^{n-1} (-1)^k x^{a+k-1} \right) dx = \int_0^1 \left(\lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} (-1)^k x^{a+k-1} \right) dx = \int_0^1 \frac{x^{a-1}}{1+x} dx$$

bo'ladi.

Demak,

$$\int_0^1 \frac{x^{a-1}}{1+x} dx = \lim_{n \rightarrow \infty} \int_0^1 \left(\sum_{k=0}^{n-1} (-1)^k x^{a+k-1} \right) dx = \sum_{k=0}^{n-1} \int_0^1 ((-1)^k x^{a+k-1}) dx = \sum_{k=0}^{n-1} \frac{(-1)^k}{a+k}. \quad (4)$$

Endi

$$\int_1^{+\infty} \frac{x^{a-1}}{1+x} dx$$

integralda $x = \frac{1}{t}$ almashtirish bajarsak, unda

$$\int_1^{+\infty} \frac{x^{a-1}}{1+x} dx = \int_0^1 \frac{t^{-a}}{1+t} dt = \int_0^1 \frac{t^{(1-a)-1}}{1+t} dt$$

bo'lib, yuqoridagi (4) munosabatga ko'ra

$$\int_1^{+\infty} \frac{x^{a-1}}{1+x} dx = \sum_{k=1}^{\infty} \frac{(-1)^k}{a-k} \quad (5)$$

bo'ladi. (3), (4) va (5) munosabatlardan

$$\int_1^{+\infty} \frac{x^{a-1}}{1+x} dx = \frac{1}{a} + \sum_{k=1}^{\infty} (-1)^k \left[\frac{1}{a+k} + \frac{1}{a-k} \right]$$

bo'lishi kelib chiqadi.

Ma'lumki,

$$\frac{1}{a} + \sum_{k=1}^{\infty} (-1)^k \left[\frac{1}{a+k} + \frac{1}{a-k} \right] = \frac{\pi}{\sin a\pi} \quad (0 < a < 1)$$

(qaralsin, 76-ma'ruza).

Demak,

$$\int_0^{+\infty} \frac{x^{a-1}}{1+x} dx = \frac{\pi}{\sin a\pi} \quad (0 < a < 1)$$

bo'ladi.

4°. $\int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx$ **integralni hisoblash.** Bunda $f(x)$ funksiya $[0, +\infty)$ da

uzluksiz, istalgan $A > 0$ da $\int_A^{+\infty} \frac{f(x)}{x} dx$ integral yaqinlashuvchi va $a > 0, b > 0$.

Berilgan integralni quyidagi ikkita integralning limiti deb qaraymiz.

$$\int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = \lim_{\delta \rightarrow 0} \left[\int_{\delta}^{+\infty} \frac{f(ax)}{x} dx - \int_{\delta}^{+\infty} \frac{f(bx)}{x} dx \right].$$

Bu tenglikning o'ng tomonidagi birinchi integralda $ax = t$, ikkinchi integralda $bx = t$ almashtirishlarni bajarib topamiz:

$$\int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = \lim_{\delta \rightarrow 0} \int_{a\delta}^{+\infty} \frac{f(t)}{t} dt - \lim_{\delta \rightarrow 0} \int_{b\delta}^{+\infty} \frac{f(t)}{t} dt = \lim_{\delta \rightarrow 0} \int_{a\delta}^{b\delta} \frac{f(t)}{t} dt.$$

Ravshanki, $f(t)$ uzluksiz funksiya, $\frac{1}{t}$ funksiya esa ishora saqlaydi (chunki $a > 0$, $b > 0$, $x \in (0, +\infty)$). Demak,

$$\int_{a\delta}^{b\delta} \frac{f(t)}{t} dt$$

integralda o'rta qiymat haqidagi teoremani qo'llash mumkin:

$$\int_{a\delta}^{b\delta} \frac{f(t)}{t} dt = f(\xi) \int_{a\delta}^{b\delta} \frac{dt}{t}.$$

Natijada,

$$\lim_{\delta \rightarrow 0} \int_{a\delta}^{b\delta} \frac{f(t)}{t} dt = \lim_{\delta \rightarrow 0} f(\xi) \int_{a\delta}^{b\delta} \frac{1}{t} dt = \lim_{\delta \rightarrow 0} f(\xi) \ln \frac{b}{a} \quad (7)$$

bo'ladi. Modomiki, ξ nuqta $a\delta$ bilan $b\delta$ orasida ekan, $\delta \rightarrow 0$ da $\xi \rightarrow 0$ va

$$\lim_{\xi \rightarrow 0} f(\xi) = f(0) \quad (8)$$

bo'ladi.

(6), (7) va (8) munosabatlardan

$$\int_0^{+\infty} \frac{f(ax) - f(bx)}{x} dx = f(0) \ln \frac{b}{a}$$

bo'lishi kelib chiqadi.

5°. Ba'zi xosmas integrallarning qiymatlari. Quyida ba'zi xosmas integrallarning qiymatlarini keltiramiz:

$$1. \int_0^{+\infty} e^{-x^2} dx = \frac{\sqrt{\pi}}{2},$$

$$2. \int_0^{+\infty} \sin x^2 dx = \frac{\sqrt{\frac{\pi}{2}}}{2},$$

$$3. \int_0^{+\infty} \cos x^2 dx = \frac{\sqrt{\frac{\pi}{2}}}{2},$$

$$4. \int_0^{+\infty} \frac{\cos xy}{1+x^2} dx = \frac{\pi}{2} e^{-|y|},$$

$$5. \int_0^{+\infty} \frac{x \sin xy}{1+x^2} dx = \frac{\pi}{2} \operatorname{sign} y e^{-|y|},$$

$$6. \int_0^{+\infty} \frac{\sin x^2}{x} dx = \frac{\pi}{4},$$

$$7. \int_0^{+\infty} \frac{\sin^4 ax - \sin^4 bx}{x} dx = \frac{3}{8} \ln \frac{a}{b},$$

$$8. \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} \sin nx dx = \operatorname{arctg} \frac{b}{n} - \operatorname{arctg} \frac{a}{n} \quad (a>0, b>0, n \neq 0),$$

$$9. \int_0^{+\infty} \frac{e^{-ax} - e^{-bx}}{x} \cos nx dx = \frac{1}{2} \ln \frac{b^2 + n^2}{a^2 + n^2}.$$

Mashqlar

1. Ushbu

$$\int_0^{+\infty} \frac{1 - \cos xy}{x} e^{-kx} dx = \frac{1}{2} \ln \left(1 + \frac{y^2}{k^2} \right) \quad (y>0, k>0)$$

tenglik isbotlansin.

2. Ushbu

$$\int_0^{+\infty} \frac{\ln(1+a^2 x^2)}{b^2+x^2} dx, \quad (a>0, b>0)$$

integral hisoblansin.

3. Ushbu

$$\int_0^{+\infty} \frac{dx}{x^2+a} = \frac{\pi}{2\sqrt{a}}$$

($a>0$) tenglikdan foydalanib,

$$\int_0^{+\infty} \frac{dx}{(x^2+a)^{n+1}}, \quad (n \in \mathbb{N})$$

integral hisoblansin.